

P3.24 Formaldehyde (HCHO) is produced by partial oxidation of methanol (CH_3OH). Several side reactions also occur, producing formic acid (HCOOH), CO , CO_2 , and H_2O . Suppose 100 kgmol/h methanol and 21.05 kgmol/h O_2 are fed to a reactor, where 40% of the methanol and 95% of the O_2 is converted to products. What must the reactor outlet composition be?

the fractional conversion?

P4.10 Your company is interested in building a plant to make ibuprofen, using the new catalytic reaction scheme described in Example 4.1. Your job is to measure the reaction rate with different catalysts, as a first step in designing a full-scale reactor. In one experiment, you mix 134 g isobutylbenzene (IBB, $\text{C}_{10}\text{H}_{14}$) with 134 g acetic anhydride (AAn, $\text{C}_4\text{H}_6\text{O}_3$) in a laboratory-scale batch reactor, adjust the temperature, add some catalyst, and wait 1 h. At the end of the hour you stop the reaction, collect all the material in the pot, and send it for chemical analysis. The chemist reports back that the pot contains 27 g IBB, 52 g acetic anhydride, 121 g isobuty-lacetophenone (IBA, $\text{C}_{12}\text{H}_{16}\text{O}$), and 68 g acetic acid (AAc, CH_3COOH). When you see these results you tell the chemist that the analysis is wrong. He protests. Who's right?

P4.19 Waste streams from various processes in a complex chemical facility are combined and burned for heat, because the cost of purifying and recycling the components is prohibitive. On a typical day, the combined waste stream contains 2 mol% acetone ($\text{CH}_3)_2\text{CO}$, 14 mol% acetic anhydride ($\text{CH}_3\text{CO})_2\text{O}$, 36 mol% methane CH_4 , 20 mol% ethane C_2H_6 , 13 mol% N_2 , 4 mol% O_2 , 8 mol% CO , and 3 mol% H_2S . The waste stream is fed to a furnace along with 20% excess air and combusted. (For combustion reactions, excess air is calculated on the assumption that the only products of combustion are CO_2 , H_2O , N_2 , and SO_2 .) Analysis of the exit flue gas gives a CO_2/CO mass ratio of 4:1. The only other components in the flue gas are N_2 , O_2 , H_2O , and SO_2 . Calculate the composition of the exit flue gas. Assume air is 21 mol% O_2 and 79 mol% N_2 .

P4.20 Your neighbors are concerned about the

CHE 210 HW #6

1. P3.24 partial oxidation of formaldehyde
 $\text{CH}_3\text{OH} + \frac{1}{2}\text{O}_2 \rightarrow \text{HCHO} + \text{H}_2\text{O}$

100 kmol/h (methanol) $\rightarrow$ feed into reactor
 21.05 kmol/h (O_2)

40% methanol $\rightarrow$ converted into products
 95% of O_2

$$40\% (\text{CH}_3\text{OH}) \rightarrow (0.40)(100) = 40 \text{ kmol/h}$$

$$95\% (\text{O}_2) \rightarrow (0.95)(21.05) \approx 20 \text{ kmol/h}$$

$$\text{O}_2 \text{ left} = 21.05 \text{ kmol/h} - 20 \text{ kmol/h} = 1.05 \text{ kmol/h}$$

Problem asks for outlet concentration, so all components. Only oxygen is here. -15 pts

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3. P4.10

			Molar Mass	Initial Moles	final mol
Isobutylbenzene	IBB	$\text{C}_{10}\text{H}_{14}$	134.22 g/mol	0.998 mol	0.201 mol
Acetic Anhydride	AAAn	$\text{C}_4\text{H}_6\text{O}_3$	102.09 g/mol	1.313 mol	0.509 mol
Isobutylacetophenone	IBA	$\text{C}_{12}\text{H}_{16}\text{O}$	176.26 g/mol	-	0.687 mol
Acetic Acid	AAc	CH_3COOH	60.05 g/mol	-	1.132 mol

134g IBB
 134g AAAn



Pot (1hr)

27g IBB
 52g AAAn
 121g IBA
 68g AAc

$$\text{Total Mass}_{\text{initial}} = 134\text{g} + 134\text{g} = 268\text{g}$$

$$\text{Final Mass} = 27\text{g} + 52\text{g} + 121\text{g} + 68\text{g} = 268\text{g}$$

$\rightarrow$ Mass conserved

If the reaction is 1:1:1:1, ξ

$$\text{IBB} = \text{IBB}_0 - \xi \quad \xi = 0.998 - 0.201 = 0.797 \text{ mol}$$

$$\text{AAAn} = \text{AAAn}_0 - \xi \quad \xi = 1.313 - 0.509 = 0.804 \text{ mol}$$

$$\text{IBA} = 0 + \xi \quad \xi = 0.687 \text{ mol}$$

$$\text{AAc} = 0 + \xi \quad \xi = 1.132 \text{ mol}$$

All aren't equal, therefore the composition isn't consistent, and therefore the analyst is wrong the #'s don't correspond to a single chemical extent consistent with the stoichiometry.

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P4.19

Acetone (CH_3COCH_3) $\rightarrow$ 2 mol

Acetic Anhydride ($\text{C}_4\text{H}_6\text{O}_3$) $\rightarrow$ 14 mol

Methane (CH_4) $\rightarrow$ 36 mol

Ethane (C_2H_6) $\rightarrow$ 20 mol

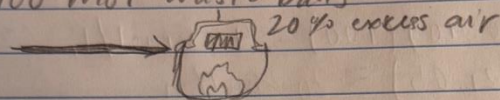
$\text{N}_2 \rightarrow$ 13 mol

$\text{O}_2 \rightarrow$ 4 mol

$\text{CO} \rightarrow$ 8 mol

$\text{H}_2\text{S} \rightarrow$ 3 mol

Per 100 mol waste basis



CO_2/CO Ratio 4:1

Air Composition : 21% O_2 , 79% N_2

$\text{C}_x\text{H}_y\text{O}_z \rightarrow \text{CO}_2 + \text{H}_2\text{O}$

$\text{O}_2 \text{ required} = x + y/4 - z/2$

Acetone ($\text{C}_3\text{H}_6\text{O}$) : 2 mol (4.0) = 8.0

Acetic Anhydride ($\text{C}_4\text{H}_6\text{O}_3$) : 14 (4.0) = 56

CH_4 : 36 (2.0) = 72

C_2H_6 : 20 (3.5) = 70

CO : 8 (0.5) = 4

H_2S : 3 (1.5) = 4.5

$\text{O}_2 \text{ Required} = 8 + 56 + 72 + 70 + 4 + 4.5 = 214.5 \text{ mol } \text{O}_2$

$214.5 - 4 = 210.5 \text{ mol } \text{O}_2$

Supplied $\text{O}_2 = (1.2)(210.5) = 252.6 \text{ mol } \text{O}_2$

Air Mols = $\frac{252.6}{0.21}$

$\text{N}_2 \text{ from Air} = 1202.86 (0.79)$

= 950.26

= 1202.86

Add feed N_2 (N_2 in exit) = $950.26 + 13 = 963.26 \text{ mol}$

$$\text{Carbon Atoms } 2 \times 3 = 6$$

$$\text{AAH: } (14)(4) = 56$$

$$\text{CH}_4: (36)(1) = 36$$

$$\text{C}_2\text{H}_6: (20)(2) = 40$$

$$\text{CO: } (8)(1) = 8$$

$$146 \text{ mol C}$$

$$\text{Hydrogen Atoms } 2 \times 6 = 12$$

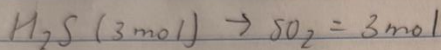
$$(14)(6) = 84$$

$$(36)(4) = 144$$

$$(20)(6) = 120$$

$$(3)(2) = 6$$

$$366/2 = 183 \text{ mol H}_2\text{O}$$



$$\text{CO moles} = C \quad \text{CO}_2 \text{ moles} = \text{total C} - C = 146 - C$$

$$\frac{44.0(146 - C)}{28.01C} = 4$$

$$44.01(146 - C) = 112.04C$$

$$6424. + (\text{approx}) - 44.01C = 112.04C$$

$$6424.4 \approx 156.05C$$

$$C \approx 41.18 \text{ mol}$$

$$\text{CO}_2 \text{ exit} = 146 - 41.18 \approx 104.82 \text{ mol}$$

Oxygen atoms

$$\text{AAH: } (14)(3) = 42$$

$$\text{CO: } 8(1)$$

$$\text{O}_2: 4 \text{ mol}(2) = 8$$

$$\text{acetone: } 2(1) = 2$$

60 mol O atoms

$$\text{O atoms: } 252.6 \text{ mol O}_2 \times 2 = 505.2 \text{ mol O atoms}$$

O₂ in products

$$\text{CO}_2: 2(104.82) = 209.64$$

$$\text{H}_2\text{O: } 1(183) = 183$$

$$\text{SO}_2: 2(3) = 6.6$$

$$\text{CO: } 1(41.18) = 41.18$$

439.82 mol O atoms

$$\text{Remaining O atoms} = 505.2 - 439.82 = 125.38 \text{ mol O atoms}$$

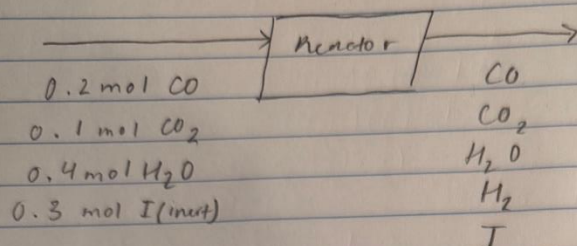
$$\text{O}_2 \text{ exit} = 125.38/2 = 62.69$$

	mole fraction	
$N_2 = 963.28$	0.7093	70.93 % ✓
$O_2 = 62.69$	0.07719	7.72 %
$CO_2 = 104.82$	0.03033	3.03 %
$CO = 41.18$	0.13476	13.48 %
$H_2O = 183$	0.04618	4.62 %
$SO_2 = 3$	0.00221	0.22 %
Sum 1357.95		100 %

Some of your numbers are labeled the wrong thing but are the correct value for a different species, so I assumed you just mixed them up and gave full credit.

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2.



DOF 4a. 4+5=9 5a 1
 4b. 1 5b 3
 5c 0
 10-1=1 5d 5

Temperature is given = 0 = DOF

$$x_{CO} = \frac{n_{CO}}{n_{total}} \quad x_{H_2} = \frac{n_{H_2}}{n_{total}}$$

$$x_{CO_2} = \frac{n_{CO_2}}{n_{total}} \quad x_{H_2O} = \frac{n_{H_2O}}{n_{total}}$$

$$n_{i,out} = n_{i,in} + \sum_{all k} \nu_{i,k} \xi_k$$

$$\begin{aligned} n_{CO,out} &= 0.2 - (1)\xi \\ n_{CO_2,out} &= 0.1 + (1)\xi \\ n_{H_2O,out} &= 0.4 + (1)\xi \\ n_{H_2,out} &= 0 + \xi \\ n_I &= 0.5 \\ n_{total} &= 1 \end{aligned}$$

$$\left(\frac{n_{CO_2}}{n_{total}} \right) \left(\frac{n_{H_2}}{n_{total}} \right) = \frac{(0.1+\xi)(\xi)}{1}$$

$$\left(\frac{n_{CO}}{n_{total}} \right) \left(\frac{n_{H_2O}}{n_{total}} \right) = \frac{(0.2\xi)(0.4\xi)}{1} = a \exp\left(\frac{b}{T}\right)$$

$$\frac{0.1 + (\xi)^2}{0.08 - 0.4\xi - 0.2\xi + \xi^2} = 0.8858$$

$$0 = -0.115\xi^2 - 0.631\xi + 0.0708$$

$$n_{H_2} = 0.11 \text{ mol}$$

$$x_{H_2} = \frac{\xi}{1} = 0.11$$

$$\xi = -5.59, (0.11)$$

c. The calculations assume its at equilibrium, which might not be accurate immediately after the setup reaction.

No numerical method, -5 pts